TABELA: Derivadas, Integrais e Identidades Trigonométricas

• *Derivadas*

Sejam u e v funções deriváveis de x e n constante.

1. $y = u^n \;\Rightarrow y' = n\,u^{n-1}u'$.
2. $y = uv \;\Rightarrow y' = u'v + v'u$.
3. $y = \frac{u}{v} \;\Rightarrow y' = \frac{u'v - v'u}{v^2}$.
4. $y = a^u \;\Rightarrow y' = a^u(\ln a)\,u',\ (a > 0,\ a \neq 1)$.
5. $y = e^u \;\Rightarrow y' = e^u u'$.
6. $y = \log_a u \;\Rightarrow y' = \frac{u'}{u}\log_a e$.
7. $y = \ln u \;\Rightarrow y' = \frac{1}{u}u'$.
8. $y = u^v \;\Rightarrow y' = v\,u^{v-1}\,u' + u^v(\ln u)\,v'$.
9. $y = \text{sen}\,u \;\Rightarrow y' = u'\cos\,u$.
10. $y = \cos\,u \;\Rightarrow y' = -u'\text{sen}\,u$.
11. $y = \text{tg}\,u \;\Rightarrow y' = u'\sec^2 u$.
12. $y = \text{cotg}\,u \;\Rightarrow y' = -u'\text{cosec}^2 u$.
13. $y = \sec\,u \;\Rightarrow y' = u'\sec\,u\,\text{tg}\,u$.
14. $y = \text{cosec}\,u \;\Rightarrow y' = -u'\text{cosec}\,u\,\text{cotg}\,u$.
15. $y = arc\,\text{sen}\,u \;\Rightarrow y' = \frac{u'}{\sqrt{1-u^2}}$.
16. $y = arc\,\cos\,u \;\Rightarrow y' = \frac{-u'}{\sqrt{1-u^2}}$.
17. $y = arc\,\text{tg}\,u \;\Rightarrow y' = \frac{u'}{1+u^2}$.
18. $y = arc\,\cot g\,u \;\Rightarrow \frac{-u'}{1+u^2}$.
19. $y = arc\,\sec\,u,\ |u| \geqslant 1$
    $\Rightarrow y' = \frac{u'}{|u|\sqrt{u^2-1}}, |u| > 1$.
20. $y = arc\,\text{cosec}\,u,\ |u| \geqslant 1$
    $\Rightarrow y' = \frac{-u'}{|u|\sqrt{u^2-1}}, |u| > 1$.

• *Identidades Trigonométricas*

1. $\text{sen}^2 x + \cos^2 x = 1$.
2. $1 + \text{tg}^2 x = \sec^2 x$.
3. $1 + \text{cotg}^2 x = \text{cosec}^2 x$.
4. $\text{sen}^2 x = \frac{1-\cos\,2x}{2}$.
5. $\cos^2 x = \frac{1+\cos\,2x}{2}$.
6. $\text{sen}\,2x = 2\,\text{sen}\,x\,\cos\,x$.
7. $2\,\text{sen}\,x\,\cos\,y = \text{sen}\,(x-y) + sen\,(x+y)$.
8. $2\,\text{sen}\,x\,\text{sen}\,y = \cos\,(x-y) - \cos\,(x+y)$.
9. $2\,\cos\,x\,\cos\,y = \cos\,(x-y) + \cos\,(x+y)$.
10. $1 \pm \text{sen}\,x = 1 \pm \cos\left(\frac{\pi}{2} - x\right)$.

• *Integrais*

1. $\int du = u + c$.
2. $\int u^n du = \frac{u^{n+1}}{n+1} + c,\ n \neq -1$.
3. $\int \frac{du}{u} = \ln|u| + c$.
4. $\int a^u du = \frac{a^u}{\ln\,a} + c,\ a > 0,\ a \neq 1$.
5. $\int e^u du = e^u + c$.
6. $\int \text{sen}\,u\,du = -\cos\,u + c$.
7. $\int \cos\,u\,du = \text{sen}\,u + c$.
8. $\int \text{tg}\,u\,du = \ln|\sec\,u| + c$.
9. $\int \text{cotg}\,u\,du = \ln|\text{sen}\,u| + c$.
10. $\int \sec\,u\,du = \ln|\sec\,u + \text{tg}\,u| + c$.
11. $\int \text{cosec}\,u\,du = \ln|\text{cosec}\,u - \text{cotg}\,u| + c$.
12. $\int \sec\,u\,\text{tg}\,u\,du = \sec\,u + c$.
13. $\int \text{cosec}\,u\,\text{cotg}\,u\,du = -\text{cosec}\,u + c$.
14. $\int \sec^2 u\,du = \text{tg}\,u + c$.
15. $\int \text{cosec}^2 u\,du = -\text{cotg}\,u + c$.
16. $\int \frac{du}{u^2+a^2} = \frac{1}{a}arc\,\text{tg}\frac{u}{a} + c$.
17. $\int \frac{du}{u^2-a^2} = \frac{1}{2a}\ln\left|\frac{u-a}{u+a}\right| + c,\ u^2 > a^2$.
18. $\int \frac{du}{\sqrt{u^2+a^2}} = \ln\left|u + \sqrt{u^2+a^2}\right| + c$.
19. $\int \frac{du}{\sqrt{u^2-a^2}} = \ln\left|u + \sqrt{u^2-a^2}\right| + c$.
20. $\int \frac{du}{\sqrt{a^2-u^2}} = arc\,\text{sen}\frac{u}{a} + c,\ u^2 < a^2$.
21. $\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a}arc\,\sec\left|\frac{u}{a}\right| + c$.

• *Fórmulas de Recorrência*

1. $\int sen^n au\,du = -\frac{sen^{n-1}au\,\cos\,au}{an} + \left(\frac{n-1}{n}\right)\int sen^{n-2}au\,du$.
2. $\int \cos^n au\,du = \frac{sen\,au\,\cos^{n-1}au}{an} + \left(\frac{n-1}{n}\right)\int \cos^{n-2}au\,du$.
3. $\int tg^n au\,du = \frac{tg^{n-1}au}{a(n-1)} - \int tg^{n-2}au\,du$.
4. $\int cotg^n au\,du = -\frac{cotg^{n-1}au}{a(n-1)} - \int cotg^{n-2}au\,du$.
5. $\int \sec^n au\,du = \frac{\sec^{n-2}au\,tg\,au}{a(n-1)} + \left(\frac{n-2}{n-1}\right)\int \sec^{n-2}au\,du$.
6. $\int cosec^n au\,du = -\frac{cosec^{n-2}au\,cotg\,au}{a(n-1)} + \left(\frac{n-2}{n-1}\right)\int cosec^{n-2}au\,du$.

Tabela: Transformada de Laplace

1	$\dfrac{1}{s}$
e^{at}	$\dfrac{1}{s-a}$
t^n	$\dfrac{n!}{s^{n+1}}$
$\operatorname{sen} at$	$\dfrac{a}{s^2+a^2}$
$\cos at$	$\dfrac{s}{s^2+a^2}$
$\operatorname{senh}\ at$	$\dfrac{a}{s^2-a^2}$
$\cosh\ at$	$\dfrac{s}{s^2-a^2}$
$e^{at}\operatorname{sen} bt$	$\dfrac{b}{(s-a)^2+b^2}$
$e^{at}\cos bt$	$\dfrac{s-a}{(s-a)^2+b^2}$
$t^n e^{at}$	$\dfrac{n!}{(s-a)^{n+1}}$
$U_c(t)$	$\dfrac{e^{-cs}}{s}$
$U_c(t)f(t-c)$	$e^{-cs}F(s)$
$e^{ct}f(t)$	$F(s-c)$
$f(ct)$	$\dfrac{1}{c}F\left(\dfrac{s}{c}\right)$
$\displaystyle\int_0^t f(t-\tau)g(\tau)d\tau$	$F(s)G(s)$
$\delta(t-c)$	e^{-cs}
$f^{(n)}(t)$	$s^nF(s)-s^{(n-1)}f(0)-\ldots-f^{(n-1)}(0)$
$(-t)^nf(t)$	$F^{(n)}(s)$